

Wenbo Yu

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RESEARCH INTERESTS

Robot learning, VLA-RL, Sim-to-real transfer, Computer Vision, Continual Learning, 3D Vision, HCI

EDUCATION

Beijing Forestry University, Beijing, China

Bachelor of Science:Automation

Sep. 2023 — Jun. 2027

Cumulative GPA: GPA: 3.82/4.00

RESEARCH EXPERIENCE

Gaoling School of AI , Renmin University of China

Research Assistant (Advisor: Prof. Di Hu)

Beijing, China

Jan. 2025 — Present

- Led Research on Sim-to-Real Transfer: Proposed GeCo-SRT, a geometry-aware continual learning framework designed to bridge the simulation-to-reality gap across sequential manipulation tasks.
- Algorithm Development: Engineered a Geometry-aware Mixture-of-Experts (Geo-MoE) module to extract invariant geometric features (e.g., planarity, linearity) and a Geo-PER replay strategy to mitigate catastrophic forgetting.
- Key Achievements: Validated the method on real-world robotic tasks, achieving a 52 % average performance improvement over baselines and reducing data requirements for new tasks to 16.7% compared to training from scratch. The paper is currently accepted by CVPR2026.
- Collaboration & Mentorship: Contributed to various robotics projects and actively mentored junior students, guiding them through experimental workflows and robot learning fundamentals.

Beijing Innovation Center of Humanoid Robotics Co., Ltd

Research Intern

Beijing, China

Dec. 2025 – Present

- **Project Lead — Real-Time Asynchronous VLA Execution Framework:** Led the development of a training-free, real-time inference plug-in for flow-based VLA models to resolve execution latency and chunk stuttering.
- Formulated the action chunk alignment problem as a barrier-guided action-space constraint, decoupling inference and control frequencies to yield highly smooth, continuous robotic manipulation with negligible computational overhead.
- **Core Contributor — Vision-Language-Action Reinforcement Learning (VLA-RL):** Drove the end-to-end development cycle of VLA-RL policies for long-horizon manipulation, encompassing task specification, custom RL reward engineering, and the supervision of large-scale teleoperation data collection.
- Deployed trained policies onto physical robotic systems via iterative sim-to-real tuning, achieving robust, millimeter-level precision manipulation that was prominently featured by CCTV and other major media outlets.

Rutgers University

Summer Research Intern (Advisor: Prof. Jingjin Yu)

NJ, USA

May. 2026 — Present

- Supported the evaluation of Affordance2Action (A2A), a framework designed to ground task-conditioned instructions to functional object parts in complex scenes.
- Conducted extensive experiments to benchmark generic segmentation and Vision-Language Model (VLM) baselines, highlighting critical gaps in existing affordance datasets.

PUBLICATIONS

- **Wenbo Yu***, Wenke Xia*, Weitao Zhang, Di Hu. GeCo-SRT: Geometry-aware Continual Adaptation for Cross-Task Sim-to-Real Transfer. 2026. (**CVPR2026**)
- **Wenbo Yu**, Jifan Li, Wenke Xia, Yuchen Li, Yizhuo Zhang, Pei Ren, Zhengping Che, Di Hu. Barrier-Constrained Guidance For Real-time Asynchronous VLA Execution. 2026. (**CoRL 2026**)
- Wenke Xia*, Pei Ren*, **Wenbo Yu**, Yizhuo Zhang, Jifan Li, Yixue Zhang, Yinuo Zhao, Qingyang Gao, Jianlong Fu, Jian Tang, Zhengping Che, Di Hu. Not All Actions are Equal: Reliable Value Estimation for Offline-to-Online Reinforcement Learning. 2026. (**CoRL 2026**)
- Litao Liu, Yifan Han, Pengfei Yi, **Wenbo Yu**, Hanqing Wang, Haoran Du, Enze Yuan, Zilin Yuan, Ruiding Feng, Michael Liu, Qi Zhang, Jingjin Yu. Affordance2Action: Task-Conditioned Scene-level Affordance Grounding for Real-Time Manipulation. 2026. (Under Review)

Academic Services

- Conference Reviewer
2026 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)
10th Annual Conference on Robot Learning (CoRL 2026)
- Student Volunteer
ACM CHI Conference on Human Factors in Computing Systems (CHI 2025)

PROJECTS

Wildlife Protection with AI

Core Team Member(Advisor: Prof. Junguo Zhang)

Beijing, China
May 2024 — Dec 2024

- Contributed to the optimization of deep learning models using Python and C++, significantly enhancing the classification accuracy and tracking efficiency of wildlife in infrared imagery.
- Conducted data preprocessing and curation on large-scale field datasets, providing a robust foundation for model training and subsequent behavioral analysis.
- Deployed infrared monitoring equipment in the field to ensure continuous, high-quality data acquisition, and presented technical reports demonstrating the project's innovation, securing the **Second Prize** in the Beijing Division of the China International College Students' Innovation Competition 2024.

AWARDS

China International College Students' Innovation Competition
Second Prize (Beijing Division)

Beijing, China
2024

"TI Cup" Beijing College Student Electronic Design Competition
Third Prize

Beijing, China
2025 & 2024

Beijing Forestry University
First Class Scholarship for Outstanding Students(Only for top 1%)

Beijing, China
2024

Beijing Forestry University
Science and Technology Innovation Scholarship

Beijing, China
2024

SKILLS

- **English:** CET-6: 559 (Pass: 450)
- **Programming:** Python, C++, C
- **Software:** Keil uVision, multisim, ROS, Microsoft SQL Server, Code Composer Studio(TI), Solid Edge, Matplotlib, LaTeX